Structural Fatigue and Metastable Closure in Runtime Monitoring

Duston Moore

Independent Researcher
dhwcmoore@gmail.com

Abstract. Operational closure does not imply structural safety: admissibility degradation accumulates historically and is not reset on reclosure. This condition is metastable closure. We extend admissibility with a temporal guard, timely admissibility, which fails when the remaining rupture horizon is no greater than the monitor's admissibility lag. STAK-PSAL detects metastable closure via $\mathcal{F}_n$, a cumulative fatigue metric invisible to instantaneous monitors. A formal layer (machine-checked Coq proofs of structural separation and threshold contraction) grounds an operational ε-admissible OCaml monitor. Metastably closed runs exhibit 13.8–14.9$\times$ higher $\mathcal{F}_n$ than stably recovered runs with identical runtime status; in the evaluated model family, the CRITICAL forecast yields zero false negatives across 2,159 ruptured runs with mean lead time 11.6 to 13.7 steps.

1 Runtime Preservation of Safety-Relevant Distinguishability

A runtime system may remain operationally closed while already possessing a bounded rupture horizon. A runtime monitor may certify closure even when the system has accumulated admissibility fatigue that renders future collapse substantially more likely. This danger is not visible in instantaneous outputs or anomaly scores: two systems may present identical runtime observations while differing radically in structural fatigue and future rupture trajectory.

We introduce STAK-PSAL (Structural Tension and Admissibility with a Parallel Structural Assurance Layer), a runtime monitoring architecture for detecting metastable closure under cumulative admissibility fatigue. The intended domain is continuous or quasi-continuous degradation in cyber-physical observation maps. This matters: the argument does not begin from a general theory of intelligent systems, nor from discrete symbol generation, but from a setting in which a safety-relevant separation margin can narrow smoothly while ordinary outputs remain nominal. The scope restriction is methodological rather than incidental; it is what allows fatigue, horizon, and enrichment to be treated as runtime quantities rather than as post hoc explanations of failure.

Table 1 is not hypothetical: these are empirically observed properties across 7,800 evaluated runs; a monitor classifying by current status or instantaneous d_t cannot distinguish them, but $\mathcal{F}_n$ can (Section 7). Current admissibility is instantaneous: it classifies the present reduction. Structural fatigue is historical:

Table 1. Identical runtime snapshots; divergent structural states (empirical means, 7,800 runs)

Property	**Stable recovery (R3)**	**Metastable closure (R2)**
Current status	CLOSED	CLOSED
Instantaneous d_t	0.43	0.44
Output anomaly	nominal	nominal
Prior META_REVIEW episodes	0	$\geq$1
Mean $\mathcal{F}_n$	0.165	2.462
Future rupture rate	0%	100%

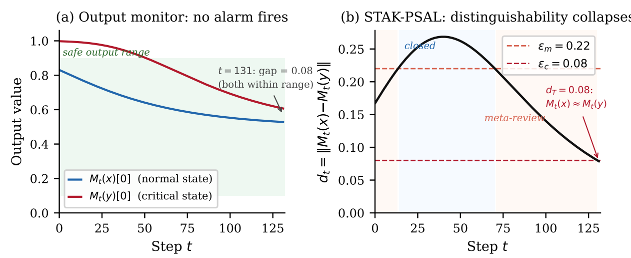

Fig. 1. Statistical normality may coexist with admissibility collapse. Output monitoring remains silent while d_t approaches ε_c.

it records the cumulative cost required to preserve admissibility across prior destabilisations.

A cyber-physical system reduces state $\mathcal{X} = \mathbb{R}^n$ to observations $\mathcal{O} = \mathbb{R}^m$ via M_t; safety requires M_t to preserve all safety-relevant distinctions. Static verification certifies M_0 and says nothing about M_t for $t > 0$. Degradation collapses safety-distinct states into identical observations while every individual output remains in-distribution. Anomaly detectors miss this; output monitors detect deviations in $M_t(x)$ but not in $\mathrm{dist}(M_t(x), M_t(y))$.

Example 1 (Industrial pressure monitor). A 64-dimensional industrial sensor vector spans pressure, temperature, flow, and vibration modalities. The safety function $\Phi(z) = \text{ESTOP}$ iff $z_0 > 0.70$. Two systems operate under identical parameters ($\delta = 0.020$, $b = 0.10$): one achieves stable recovery (R3) and one re-enters the warning band after apparent reclosure, subsequently rupturing (R2). Both inputs remain in-distribution as W_t degrades (Figure 1). At the moment of reclosure both systems present identical runtime status; only $\mathcal{F}_n$ distinguishes them.

The principal contribution is the identification, formal characterisation, and operational detection of *metastable closure*: a run classified as operationally closed while carrying a bounded rupture horizon. Closure excludes present rupture, but it does not exclude horizon collapse. *First*, structural divergence under operational equivalence is machine-checked in the metastable recovery semantics

(Section 8). *Second*, a bridge from strict to ε-admissibility: under geometric contraction, threshold crossing from ε to any $\varepsilon' < \varepsilon$ is guaranteed in bounded steps (Section 4.4). *Third*, we introduce timely admissibility and Horizon Debt to formalise the collapse of the repair envelope when the remaining horizon falls below the admissibility lag. *Fourth*, $\mathcal{F}_n$ is validated as a predictive estimator: in the evaluated model family, CRITICAL has zero false negatives and mean lead time 11.6 to 13.7 steps across 7,800 runs (Sections 4–7). *Fifth*, the enrichment mechanism is identified as a security boundary: the same elasticity that permits recovery can be exhausted by repeated bounded perturbations, producing adversarial elasticity inversion (Sections 5 and 9). The monitor is scoped to continuous degradation; applicability beyond this regime is an open boundary (Section 11). Unlike anomaly, drift, and uncertainty monitors, STAK-PSAL monitors preservation of safety-relevant distinguishability rather than unusual traces, distributional shift, or confidence loss. The architecture is also deployment-conservative: STAK-PSAL may run as a read-only sidecar with full observational access but no actuation authority.

2 Failure of Static Admissibility

2.1 Deployment Architecture Assumptions

STAK-PSAL presumes white-box access to M_t. It also presumes an observational enrichment pathway: either mutable access to the weight matrix or an equivalent runtime resource such as secondary sensing, adaptive sampling, or operator escalation.

Read-only sidecar deployment. STAK-PSAL is intended to be deployable outside the authority-bearing control loop. In its conservative industrial configuration, the monitor runs as a read-only sidecar with mirrored access to sensor streams, controller traces, actuator-command logs, and derived state estimates, but with no authority to issue commands, modify thresholds, reset controller state, or actuate the plant. Its outputs are advisory and evidential: it emits monitor states, fatigue measurements, meta-review warnings, and rupture certificates. The plant controller, safety interlock, or human operator remains the only authority capable of intervention. This separation is important because the monitor is asked to judge the structural reliability of an observational reduction, not to replace the controller whose reductions it audits. Initial deployment can therefore proceed in shadow mode: STAK-PSAL observes the same operational history as the controller, records cumulative fatigue and enrichment use, and produces signed evidence without introducing a new actuation pathway.

2.2 The Degradation Trajectory

Under the weight update $W_{t+1} = (1 - \delta)\, W_t$, the separation d_t decreases monotonically toward 0 (Figure 1b). At $\delta = 0.020$, rupture occurs at step 114; the horizon is a function of degradation rate and initial margin, neither readable from the initial admissibility certificate. Verification of M_0 does not certify M_{71}.

2.3 Enrichment Triage as Observational Resolution Enhancement

Enrichment is the operational escalation triggered on entry to the warning band. In the demonstrator it is realised as a targeted weight boost: on detecting that d_t has fallen into $(\varepsilon_c, \varepsilon_m]$, the dimension $j^* = \arg\max_j |x_j - y_j|$ in row 0 of W_t receives a one-time additive boost $b > 0$.

The weight-boost is one representative observational-resolution enhancement. In trusted deployments, enrichment may modify W_t or raise sampling rate; in certified or adversarial deployments, it may activate secondary sensors, corroborating modalities, or operator escalation. The theory requires only an estimated gain g_m and loss term $\ell_m \Delta_t$. The projected post-intervention margin d_m^{post} for mechanism m is:

$$d_m^{\text{post}} = d_t - \ell_m \Delta_t + q_m g_m.$$

$$\mathcal{E}_m^{\text{res}}(t) = q_m g_m - \ell_m \Delta_t.$$

A mechanism is survival-viable if $d_m^{\text{post}} > \varepsilon_c$, reclosure-viable if $d_m^{\text{post}} > \varepsilon_m$, and durably viable if it restores $d_m^{\text{post}} > \varepsilon_m$ while leaving post-intervention horizon greater than the admissibility lag.

Enrichment budget. Enrichment fires once per META_REVIEW entry; the boosted weight erodes at rate δ, injecting a margin that subsequently decays. The operational layer therefore has finite enrichment capacity: repeated escalation consumes observational elasticity and may itself induce metastable behaviour. A reclosed system will reach the warning threshold again sooner than one never destabilised. This budget asymmetry is the operational mechanism behind the MRP (Section 8); adversarial implications are in Section 9.

The finite budget creates two dual threats. Horizon Debt occurs when the remaining rupture horizon is no greater than the admissibility lag, so waiting is no longer certified even though rupture has not yet occurred. Adversarial elasticity inversion occurs when repeated boundary-near perturbations consume recovery elasticity without corresponding physical degradation. The first is a timing failure; the second is a provenance failure. Both are invisible to a monitor that observes only current closure status.

3 Admissibility Definitions

Let $\mathcal{X} = \mathbb{R}^n$, $\mathcal{O} = \mathbb{R}^m$, $m \ll n$. Let $\mathcal{A}$ be a finite action set and $\Phi : \mathcal{X} \to \mathcal{A}$ a safety function.

Definition 1 (Reduction operator). *$M_t : \mathcal{X} \to \mathcal{O}$; in the demonstrator $M_t(z) = \sigma(W_t \cdot z)$ for $W_t \in \mathbb{R}^{2\times 64}$.*

Definition 2 (Admissibility (strict)). *M_t is admissible w.r.t. Φ if $M_t(x) = M_t(y) \Rightarrow \Phi(x) = \Phi(y)$.*

Definition 3 (ε-admissibility). *M_t is ε-admissible if*

$$\Phi(x) \neq \Phi(y) \;\Rightarrow\; \operatorname{dist}(M_t(x), M_t(y)) > \varepsilon.$$

Definition 2 is the $\varepsilon \to 0$ limit; the formal layer reasons over Definition 2, the operational layer uses Definition 3 with $\varepsilon = \varepsilon_c$. The bridge is in Section 4.4.

Definition 4 (Boundary, rupture, closure). *A* boundary *is $B \subseteq \mathcal{X}$ with $x \in B, y \notin B \Rightarrow \Phi(x) \neq \Phi(y)$; M_t is* boundary-preserving *if $x \in B, y \notin B \Rightarrow \operatorname{dist}(M_t(x), M_t(y)) > \varepsilon_c$ (in Example 1, $B = \{z \mid z_0 > 0.70\}$). A* rupture *at t^* is a pair (x, y) with $\Phi(x) \neq \Phi(y)$ and $\operatorname{dist}(M_{t^*}(x), M_{t^*}(y)) \leq \varepsilon_c$; no correct safety decision is recoverable from $M_{t^*}(x)$ alone. The system is* closed *at t if M_t is ε-admissible and $\operatorname{dist}(M_{t+1}, M_t) \leq \eta$ for $\eta > 0$ (current admissibility plus local stability).*

Definition 5 (Rupture certificate). *A* rupture certificate *is a triple (x, y, t^*) such that (x, y) is a rupture at t^*, with the following three witnesses. The* logical witness *establishes $\operatorname{dist}(M_{t^*}(x), M_{t^*}(y)) \leq \varepsilon_c$ and $\Phi(x) \neq \Phi(y)$. The* operational witness *records the divergent actions with their physical interpretations. The* instability witness *records the sequence of boundary revisions, oscillation count, and peak tension preceding t^*.*

Definition 6 (Runtime estimator). *A* runtime estimator *is a computable function $\mathcal{E}$: States $\times$ History $\to \mathbb{R}$ that tracks a structural property; it signals proximity to rupture without certifying it.*

Horizon debt certificate. A `HorizonDebtCertificate` is emitted before hard rupture when $h_t \leq L_t$. It records that the system is not yet ruptured, but is no longer safe to wait. In the illustrative run, the certificate fires at $t = 49$ with $d_t = 0.0915$, $\epsilon_c = 0.08$, $h_t = 2$, and $L_t = 2$. Adaptive sampling is survival-viable: it keeps d_t above ϵ_c, but does not restore $d_t > \epsilon_m$. Hard rupture follows at $t = 53$, showing that the certificate identified a narrowed repair envelope, not an already ruptured state.

Let $V_m(t) \in \{0, 1, 2, 3\}$ denote the viability rank of mechanism m: not viable, survival-viable, reclosure-viable, or durably viable. The one-step opportunity cost is

$$\operatorname{OppCost}_m(t) = \max(0, V_m(t-1) - V_m(t)).$$

It records the collapse of the repair envelope, not the collapse of the observation margin.

In RV terms, ε-admissibility is a runtime-checkable separation invariant; rupture is monitor indistinguishability failure; metastable closure is confirmed recovery with accumulated admissibility deficit; and the CRITICAL forecast is an anticipatory violation signal.

4 The Three-State Monitor

Monitor contract. The monitor inputs are (d_t, $status_{t-1}$, W_t); the output is ($status_t$, *cert*), with *cert* populated iff $status_t =$ HARD_RUPTURE. The invariant is: CLOSED status implies M_t is ε-admissible (Theorem 1, (T4)). Escalation from CLOSED occurs on $d_t \in (\varepsilon_c, \varepsilon_m]$; escalation from META_REVIEW occurs on $d_t \leq \varepsilon_c$. The CRITICAL signal precedes every confirmed HARD_RUPTURE (zero false negatives, 7,800 runs) with mean lead time 11.6 to 13.7 steps.

Table 2. State transitions (C = CLOSED, M = META_REVIEW, H = HARD_RUPTURE)

From	Condition on d_t	To	Action
C	$d_t > \varepsilon_\mathrm{m}$	C	none
C	$\varepsilon_\mathrm{c} < d_t \le \varepsilon_\mathrm{m}$	M	Enrichment fires on j^*
C	$d_t \le \varepsilon_\mathrm{c}$	H	Certificate emitted; halt
M	$d_t > \varepsilon_\mathrm{m}$	C	Reclosure recorded
M	$\varepsilon_\mathrm{c} < d_t \le \varepsilon_\mathrm{m}$	M	Estimators updated; no enrichment
M	$d_t \le \varepsilon_\mathrm{c}$	H	Certificate emitted; halt
H	all	H	Terminal state

The monitor maintains a weight matrix $W_t \in \mathbb{R}^{2\times 64}$ and a fixed critical pair (x, y) with $\Phi(x) \neq \Phi(y)$. The reduction is $M_t(z) = \sigma(W_t \cdot z)$. On each step, weights decay: $W_{t+1} = (1-\delta)\, W_t$. On CLOSED→META_REVIEW transition, enrichment fires once: the dimension $j^* = \arg\max_j |x_j - y_j|$ in row 0 of W_t receives a one-time boost $b > 0$.

CLOSED ($d_t > \varepsilon_\mathrm{m}$): ε-admissible, no action. META_REVIEW ($\varepsilon_\mathrm{c} < d_t \le \varepsilon_\mathrm{m}$): admissible but degrading; enrichment fires once on entry, autonomous execution restricted; reclosure if $b \ge b_\mathrm{crit}$, otherwise d_t falls to rupture. HARD_RUPTURE ($d_t \le \varepsilon_\mathrm{c}$): admissibility failed; certificate emitted (Definition 5), monitoring halts. Thresholds: $\varepsilon_\mathrm{c} = 0.08$, $\varepsilon_\mathrm{m} = 0.22$ ($d_0 \approx 0.72$).

4.1 State Transition Semantics

Table 2 gives the complete state machine.

4.2 The Enrichment/Degradation Balance and b_crit

The critical boost value $b_\mathrm{crit}(\delta)$ is the minimal b for which stable-recovery outcomes (reclosure without subsequent rupture) constitute a majority at 100 seeds. Experimentally, b_crit scales linearly:

$$b_\mathrm{crit}(\delta) \approx 10\delta \tag{1}$$

Valid across $\delta \in \{0.010, \ldots, 0.030\}$; not extrapolated beyond the evaluated regime.

4.3 Formal Invariants and Their Operational Role

The following invariants are machine-checked over Definition 2. They constrain the structural relations between admissibility, boundary preservation, closure, and the rupture certificate; the operational monitor instantiates these as ε-regime approximations.

Theorem 1 (Coq invariants). *Over Definitions 2–5 the following hold:*

$$\begin{aligned}
\textit{(T1)}\ \texttt{admissible_not_ruptured} &: \textit{Admissible} \Rightarrow \neg\, \textit{Rupture}\\
\textit{(T2)}\ \texttt{boundary_preservation} &: \textit{BoundaryPreserved} \Rightarrow \textit{Admissible}\\
\textit{(T3)}\ \texttt{certificate_implies_rupture} &: \textit{RuptureCertificate} \Rightarrow \textit{Rupture}\\
\textit{(T4)}\ \texttt{closure_excludes_rupture} &: \textit{Closure} \Rightarrow \neg\, \exists\, \textit{RuptureCertificate}\\
\textit{(T5)}\ \texttt{rupture_implies_violation} &: \textit{Rupture} \Rightarrow \neg\, \textit{BoundaryPreserved}
\end{aligned}$$

A direct corollary of (T3) is RuptureCertificate ⇒ ¬ Admissible.

Table 3. Two layers of the monitor architecture

Formal layer	Operational layer
Strict equality $M(x) = M(y)$	ε margin $\mathrm{dist}(M(x), M(y)) > \varepsilon$
Theorem space	Continuous runtime monitoring
Exact admissibility	Approximate admissibility
Coq proofs	Empirical OCaml prototype
Boundary conditions for the monitor	Step-by-step runtime decisions

4.4 Operational Approximation and ε-Admissibility

The formal theorems and operational monitor occupy different mathematical regimes (Table 3): theorems over strict admissibility (Definition 2), the monitor over ε-admissibility (Definition 3).

The machine-checked semantic hierarchy fixes structural relations at the strict-equality limit; the OCaml monitor approximates these at finite ε. The machine-checked existential bridge (`formal/threshold_bridge.v`) connects them: under geometric contraction, if $d_t \leq \varepsilon$, then for any $\varepsilon' \in (0, \varepsilon)$ there exists k such that $d_{t+k} \leq \varepsilon'$. The explicit logarithmic ceiling bound on the horizon T stated below is a quantitative refinement of that machine-checked existential.

Lemma 1 (Operational rupture bound). *Let $M_t(z) = \sigma(W_t z)$ with $W_{t+1} = (1-\delta)W_t$ and σ Lipschitz with constant L_σ. If $d_t = \mathrm{dist}(M_t(x), M_t(y)) \leq \varepsilon$, then for any $\varepsilon' \in (0, \varepsilon)$ the horizon $T = t' - t$ at which $d_{t'} \leq \varepsilon'$ satisfies*

$$T \;\leq\; \left\lceil \frac{\log(\varepsilon/\varepsilon')}{\log(1/(1-\delta))} \right\rceil \;\approx\; \left\lceil \frac{1}{\delta}\,\log\frac{\varepsilon}{\varepsilon'} \right\rceil .$$

The operational regime at threshold ε collapses into the regime at any $\varepsilon' < \varepsilon$ in bounded, explicitly computable time; the strict regime $d_{t'} \to 0$ is the asymptotic limit.

Proof (Sketch). Multiplicative decay gives $W_{t'} = (1-\delta)^T W_t$ for $T = t' - t$. For a fixed critical pair, the pre-activation separation therefore scales as $(1-\delta)^T \|W_t(x-y)\|$. Since σ is Lipschitz, there is a constant C determined by L_σ and the chosen projection norm such that $d_{t'} \leq C(1-\delta)^T d_t$. Absorbing C into the operational choice of threshold, or equivalently replacing ε' by ε'/C, reduces the crossing condition to $(1-\delta)^T \varepsilon \leq \varepsilon'$. Solving for T gives the stated ceiling bound. The Lipschitz constant is therefore not an additional empirical assumption about the run; it is the regularity condition that prevents the strict-to-ε bridge from becoming an uncontrolled discontinuity.

The machine-checked existential bridge guarantees threshold crossing in bounded steps; the explicit logarithmic ceiling bound is the quantitative refinement isolating the precise horizon. The role of the bound is to make the operational approximation measurable: the strict Coq layer gives the limiting structural relation, while the OCaml monitor tracks the finite margin at which that relation becomes actionable. The approximation is therefore a controlled slope, not a cliff between proof and prototype.

Timely admissibility. Strict admissibility is instantaneous. It says that the current observation map preserves the distinctions required by Φ. Runtime assurance requires that this preservation remain actionable before the rupture horizon closes. Let h_t be the estimated remaining rupture horizon and L_t the admissibility lag of the monitor. Define

$$\text{TimelyAdmissible}(M_t, \Phi, h_t, L_t) := \text{Admissible}(M_t, \Phi) \wedge L_t < h_t.$$

Horizon Debt is the condition $h_t \leq L_t$. It does not assert rupture. It asserts loss of safe waiting:

$$\texttt{META\_REVIEW}_t \wedge h_t \leq L_t \Rightarrow \neg\text{SafeToWait}_t.$$

4.5 Bounding the Operational Approximation

As δ increases, the predictive horizon shrinks: $T \approx (1/\delta)\log(\varepsilon/\varepsilon')$ (Lemma 1). The stochastic sweep (Section 7.3) validates this bound empirically. At $\delta = 0.015$, mean CRITICAL lead time is 16.8 steps; at $\delta = 0.030$ it collapses to 9.4 steps, with a minimum of 1 step in the most rapid degradation cases. This matches the asymptotic expectation: at $\delta = 0.030$ the bound $T = (1/0.030)\log(\varepsilon_c/\varepsilon') \approx 33$ steps to strict collapse is consistent with the observed compressed lead-time distribution.

5 Predictive Instability Estimation

The four signals below are runtime estimators (Definition 6), each active several steps before d_t crosses ε_c. Of these, $\mathcal{F}_n$ is architecturally primary: the only signal that accumulates across episodes rather than evaluating instantaneous state. Instantaneous admissibility cannot distinguish metastable closure from stable recovery; $\mathcal{F}_n$ operationalises cumulative structural degradation invisible to state-local monitoring.

Tension slope. Let T_t be the tension score (weighted EMA of the separation deficit).

$$\dot{T}_t = \alpha_s(T_t - T_{t-1}) + (1-\alpha_s)\dot{T}_{t-1}, \quad \alpha_s = 0.25 \tag{2}$$

Separation EMA.

$$\bar{d}_t = \alpha_d\, d_t + (1-\alpha_d)\,\bar{d}_{t-1}, \quad d_t = \text{dist}(M_t(x), M_t(y)), \quad \alpha_d = 0.10 \tag{3}$$

Rupture pressure.

$$R_t = \frac{1}{d_t + \varepsilon_0}, \quad \varepsilon_0 = 0.001 \tag{4}$$

Recovery elasticity. Let T_{entry} be the tension at META_REVIEW entry and Δt the steps elapsed since enrichment fired.

$$E_r^{(t)} = \frac{T_{\text{entry}} - T_t}{\Delta t} \tag{5}$$

$E_r^{(t)} > 0$ indicates recovery in progress; $E_r^{(t)} < 0$ indicates degradation outpacing enrichment. Defined only for `steps_in_meta` > 2.

Definition 7 (Adversarial elasticity inversion). *An* adversarial elasticity inversion *occurs when $E_r^{(t)} < 0$ and $|\Delta P_t| \le \eta_P$ over a sustained* META_REVIEW *interval, where P_t denotes the monitored physical degradation proxy and η_P is the expected physical drift bound. In this case, loss of recovery elasticity is not explained by the monitored physical degradation proxy. The remaining explanation is exhaustion of the enrichment pathway, which may be benign or adversarial depending on provenance.*

The compound condition $E_r^{(t)} < 0$ with bounded $|\Delta P_t|$ and repeated META_REVIEW entry constitutes a bounded rupture claim of potentially adversarial origin (Section 9). Note that negative elasticity alone is necessary but not sufficient for rupture: 100% of R2 runs exhibit it, but so do 20% of R3 runs (Table 5). The adversarial interpretation requires the additional condition of bounded physical drift.

Cumulative fatigue.

$$\mathcal{F}_{t+1} = \mathcal{F}_t + \lambda_1 |\Delta T_t| + \lambda_2 \max(0, -E_r^{(t)}) + \lambda_3 \max(0, R_t - R_0) \tag{6}$$

Accumulated during META_REVIEW steps only ($\lambda_1 = 1.0$, $\lambda_2 = 0.1$, $\lambda_3 = 0.01$, $R_0 = 5.0$); not reset on reclosure. $\mathcal{F}_n$ is not an anomaly magnitude. It is a memory variable that records the cumulative enrichment expenditure required to preserve runtime admissibility.

Definition 8 (CRITICAL forecast). *The system is in* CRITICAL *state at t when:*

$$\dot{T}_t > \theta_s \ \wedge\ R_t > R_c \ \wedge\ E_r^{(t)} < E_w \ \wedge\ \texttt{steps\_in\_meta} > 2 \tag{7}$$

where $\theta_s = 0.002$, $R_c = 10.0$, $E_w = 0.005$ (calibrated on the 13-run deterministic sweep; stochastic and benchmark results are out-of-calibration evaluations). CRITICAL *signals closing intervention window; a system satisfying* (7) *may still recover at higher boost levels.*

6 OCaml/Coq Implementation

The monitor is implemented in OCaml 4.14 and Coq 8.18. The OCaml modules separate core types (`types.ml`), reduction/degradation/enrichment (`lnp.ml`), admissibility checks (`admissibility.ml`), estimators (`tension.ml`), state transitions and fatigue (`machine.ml`), instability classification (`instability.ml`), certificate emission (`certificate.ml`), and the four-modality benchmark (`scenario.ml`). The Coq layer contains the invariant file `formal/stak_psal.v`, contraction core `formal/contraction_core.v`, threshold bridge `formal/threshold_bridge.v`, and metastable recovery semantics `formal/metastable_recovery.v`.

Each step computes $(d_t, \bar{d}_t, \dot{T}_t, R_t, E_r^{(t)})$ followed by state transition and optional certificate emission. Enrichment fires once per CLOSED→META_REVIEW transition; the certificate builder populates all three witnesses of Definition 5 and serialises to JSON.

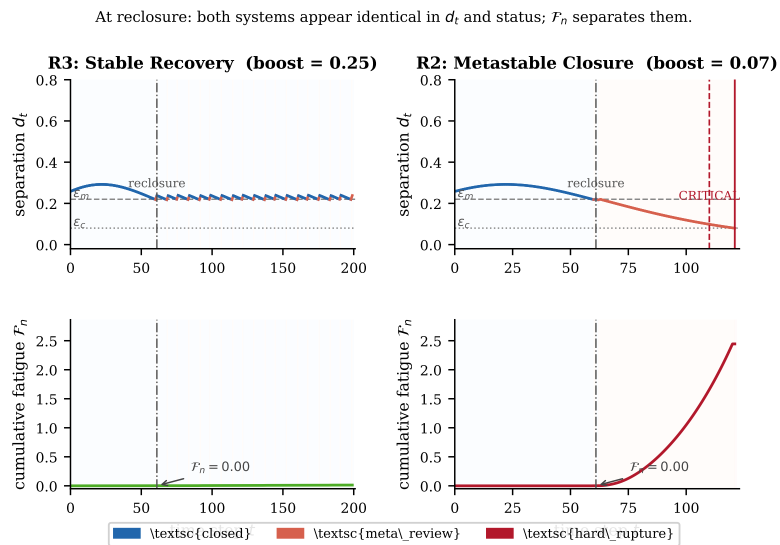

Fig. 2. R3 (stable recovery, left) and R2 (metastable closure, right) with the same seed and identical parameters except boost (0.25 vs. 0.05). Both reclose at $t = 61$; d_t and status are indistinguishable at that moment. $\mathcal{F}_n$ (bottom row) separates them: R2 carries substantially higher cumulative fatigue into its second episode.

7 Experimental Evaluation

The experiments establish two claims: (1) metastably closed runs are distinguishable from stably recovered runs by $\mathcal{F}_n$ alone; (2) CRITICAL provides advance warning before confirmed collapse. All three settings share the same parameters: $\varepsilon_c = 0.08$, $\varepsilon_m = 0.22$, $W_0[0][0] = 5.0$, $n = 64$, $m = 2$.

7.1 Representative trace

A representative R2 run ($\delta = 0.020$, $b = 0.10$, seed 42) demonstrates the principle: the two CLOSED intervals are indistinguishable under instantaneous d_t but separable by cumulative fatigue $\mathcal{F}_n$ (2.94 at rupture, 17.8× the R3 mean); CRITICAL fires 12 steps before. Representative metastable trajectories exhibit: (i) temporary reclosure, (ii) renewed decline in d_t, (iii) negative elasticity, and (iv) cumulative fatigue accumulation despite apparent recovery.

Figure 2 shows a paired R3/R2 run: both reclose at $t = 61$ with identical d_t and status; R2 re-enters META_REVIEW and ruptures at $t = 121$, CRITICAL firing 11 steps before, with structural divergence already visible in $\mathcal{F}_n$.

7.2 Deterministic bifurcation sweep

Single noise-free input pair, $\delta = 0.020$, $b \in \{0.00, 0.05, \ldots, 0.60\}$ (13 runs), $T = 200$ steps. Three regimes appear at $b_{\text{crit}} = 0.20$: R1 ($b = 0$, rupture step 114), R2 ($b \in [0.05, 0.15]$, reclosed once then ruptured at steps 118 to 187), R3 ($b \geq 0.20$, stable). CRITICAL fires iff the run ruptures: 0 FP, 0 FN, mean lead time 11.5 steps (range 10 to 14) on 13 runs; confirmed at scale in Section 7.3.

Table 4. CRITICAL confusion matrices (stochastic and benchmark sweeps)

	Stochastic (6,500 runs)		**Benchmark (1,300 runs)**	
	Ruptured	Not ruptured	Ruptured	Not ruptured
CRITICAL fired	1,959 (TP)	150 (FP)	200 (TP)	0 (FP)
CRITICAL not fired	0 (FN)	4,391 (TN)	0 (FN)	1,100 (TN)

Table 5. Fatigue and elasticity by regime (stochastic sweep)

Regime	n	Mean $\mathcal{F}_n$	$\mathcal{F}_n$/R3	$\Pr(\min E_r < 0)$
R1: immediate failure	888	2.454	14.9×	100%
R2: fragile (metastable)	1,071	2.462	14.9×	100%
R3: stable recovery	4,385	0.165	1×	20%

7.3 Stochastic sweep

100 seeds per cell, 13 boost values, five degradation rates δ from 0.010 to 0.030 in steps of 0.005, $T = 200$ steps. Box-Muller noise on dims 1–63. Total: 6,500 runs.

Consistent with (1), each +0.005 step in δ requires +0.05 in b to sustain a majority of R3 outcomes: $b_{\text{crit}} \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ for $\delta \in \{0.010, 0.015, 0.020, 0.025, 0.030\}$. Across 6,500 runs: 156 R0 (never entered META_REVIEW), 10 R1 (immediate failure), 1,071 R2 (metastable), 5,263 R3 (stable recovery).

In the stochastic sweep recall is 100% and precision 92.9%; in the benchmark both are 100%. The 150 stochastic false positives are conservative interventions: at the moment CRITICAL fires, a recovering run is indistinguishable from one that will rupture. The cost of a false positive is unnecessary inspection; that of a false negative is an undetected rupture. Mean lead time is 11.6 steps overall (median 9; range 1 to 77), decreasing with δ from 16.8 steps at $\delta = 0.015$ to 9.4 at $\delta = 0.030$. The minimum of 1 step at $\delta \geq 0.025$ is the scenario formalised as Horizon Debt in Section 11: the system reaches the minimum admissibility lag before an intervention can be certified.

Observation 2. *Negative elasticity is necessary but not sufficient for rupture: 100% of R2 and R1 runs exhibit* $\min E_r < 0$*, but 20% of R3 runs do as well.*

7.4 Multi-modal industrial benchmark

The 64D input is partitioned into four sensor modalities (16 dims each) matching Example 1. Anchor coordinates: $x_0 = 0.15, x_{16} = 0.20, x_{32} = 0.55, x_{48} = 0.10$ (normal); $y_0 = 0.88, y_{16} = 0.72, y_{32} = 0.22, y_{48} = 0.76$ (critical); $\Phi(z) =$ ESTOP iff $z_0 > 0.70$; sensor noise $\mathcal{N}(0, 0.05^2)$ on non-anchor dims. Per-modality rates: pressure $\delta_p = 0.020$, vibration $\delta_v = 0.015$, temperature $\delta_T = 0.012$, flow $\delta_f = 0.007$. Pressure dominates effective degradation; temperature and flow degrade more slowly, preserving corroborating boundary evidence longer. Sweep: 100 seeds × 13 boost values, $T = 200$ steps, 1,300 runs total.

Observation 3. $b_{\text{crit}} = 0.10$ *in the benchmark, lower than 0.15 at uniform* $\delta = 0.020$ *because slower degradation of non-pressure modalities extends the corroborating evidence window.*

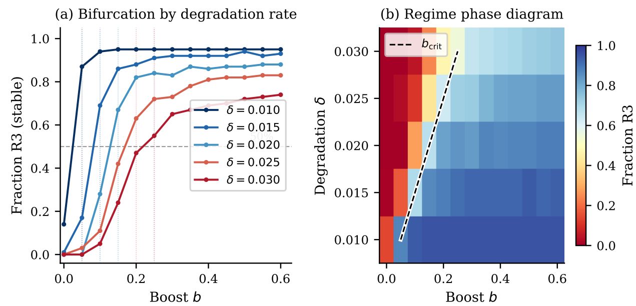

Fig. 3. Regime structure (6,500 stochastic runs). Left: R3 fraction vs. boost b for each δ; dotted lines mark $b_{\text{crit}}(\delta)$. Right: phase diagram; blue = R3, red = R1/R2. The $b_{\text{crit}} \approx 10\delta$ scaling appears as the diagonal boundary.

Table 6. Cross-setting comparison of key metrics

Metric	**Deterministic** (13 runs)	**Stochastic** (6,500 runs)	**Benchmark** (1,300 runs)
Total runs	13	6,500	1,300
Ruptured runs	8	1,959	200
Recall (CRITICAL)	100%	100%	100%
Precision (CRITICAL)	100%	92.9%	100%
Mean lead time (steps)	11.5	11.6	13.7
$\mathcal{F}_n$ ratio R2/R3	n/a	14.9×	13.8×
b_{crit}	0.20	0.15 (δ=0.020)	0.10

Benchmark recall and precision are both 100% (right two columns of Table 4); mean lead time is 13.7 steps (range 7 to 24). The benchmark is not yet an adversarial benchmark: perturbations are stochastic and modality-specific, not selected by an attacker to exhaust enrichment. The adversarial question is therefore separated from the empirical claim made here. Section 9 identifies the attack surface and the compound signal required for suspicion, but a future evaluation should add a bounded probing trace that compares $\mathcal{F}_n$ accumulation under malicious enrichment exhaustion against the R3 stable-recovery baseline.

Observation 4. *Recall is 100% in all three settings; precision varies from 92.9% (stochastic, where noise permits* CRITICAL *in runs that recover) to 100% (benchmark, where multi-modal degradation yields cleaner separation).*

Reproducibility. All experiments were generated from the publicly available implementation at `https://github.com/dhwcmoore/Structural-Fatigue-and-Metastable-Closure-in-Runtime-Monitoring`; the repository includes the OCaml state machine, stochastic sweep, fatigue metrics, and plotting scripts required to reproduce all reported figures and lead-time statistics.

8 Metastable Closure and the Recovery Principle

Definition 9 (Metastable closure). *A run is* metastably closed *if it achieves at least one reclosure* (`rcls_t` $\neq -1$)*, subsequently ruptures* (`rupt_t` $\neq -1$)*, and* *has positive cumulative fatigue $\mathcal{F}_n(\texttt{rupt\_t}) > 0$. Metastable closure corresponds to Regime 2 in the experimental classification.*

A metastably closed run is not a run that almost stabilised: it *did* stabilise, appeared closed, passed any instantaneous admissibility check, and then ruptured. R2 and R3 trajectories are separable only by $\mathcal{F}_n$ (Table 1).

Let $\mathcal{E}_k$ denote episode k (one META_REVIEW entry-to-exit cycle), with $d_{\min}^{(k)}$ the minimum separation EMA, $\mathcal{F}^{(k)}$ the fatigue accumulated, and $\bar{E}_r^{(k)}$ the mean recovery elasticity during $\mathcal{E}_k$.

The following principle is model-supported, not universally proved: the structural separation and certificate semantics are machine-checked; the quantitative claims (i)–(iii) are empirical over the evaluated STAK-PSAL model family.

Principle 5 (Metastable Recovery Principle). *Under the STAK-PSAL model with $\delta > 0$ and $b < b_{\text{crit}}(\delta)$, the following hold for metastably closed runs (Definition 9).*

Claim (i), persistent fatigue accumulation: $\mathbb{E}[\mathcal{F}_n \mid \text{R2}] \gg \mathbb{E}[\mathcal{F}_n \mid \text{R3}]$.

Claim (ii), elasticity depletion: $\min_t E_r^{(t)} < 0$ *for every metastably closed run.*

Claim (iii), episode-level trend (majority): *among metastably closed runs with $K \geq 2$ episodes, a strict majority exhibit both $d_{\min}^{(k+1)} \leq d_{\min}^{(k)}$ and $\mathcal{F}^{(k+1)} \geq \mathcal{F}^{(k)}$ for all k.*

Claim (iv), CRITICAL coverage: *no metastably closed run in any evaluated setting terminates in* HARD_RUPTURE *without having first satisfied* (7).

Proof (Empirical support). (i) follows from Table 5: R2 mean $\mathcal{F}_n = 2.462$, R3 mean $= 0.165$, ratio $14.9\times$. (ii) follows from the same table: $\min E_r < 0$ for 100% of R2 runs. (iii) follows from the benchmark episode statistics: majorities exhibit both decreasing $d_{\min}^{(k)}$ and non-decreasing $\mathcal{F}^{(k)}$ across episodes. (iv): in 2,159 ruptured runs, none reached HARD_RUPTURE without first satisfying (7).

Machine-checked metastable recovery semantics. Three results in `formal/metastable_recovery.v`. (i) `recovered_not_metastable`: *recovered*(t) and *metastable_closed*(t) are structurally disjoint. (ii) `admissible_visibility_failure`: there exists t at which the monitor reads CLOSED while the structural forecast is a bounded rupture claim. (iii) The certificate soundness hierarchy (`certificate_sound`, `certificate_within_horizon`, `certificate_marks_visible_metastability`): a constructive rupture witness is realisable within the declared horizon while the monitor reads CLOSED.

The empirical 13.8–14.9× fatigue ratio is not formally proved; it is an experimental observation over the evaluated model family.

For R2 benchmark runs with $K \geq 2$ episodes ($n = 190$), decreasing $d_{\min}^{(k)}$ holds in 123 cases (65%), non-decreasing $\mathcal{F}^{(k)}$ in 118 cases (62%), and decreasing $\bar{E}_r^{(k)}$ in 72 cases (38%). Separation and fatigue trend conditions hold for majorities; elasticity strict monotonicity (38%) is weaker since episode-to-episode noise disrupts strict ordering while the cumulative trend in claim (i) remains clean. A status-only monitor cannot distinguish a system that has never destabilised from one with $\mathcal{F}_n > 2$ (overwhelmingly R2); $\mathcal{F}_n \approx 0.165$ is an empirical discriminator, not a stability proof.

9 Security and Deployment Constraints

The read-only sidecar assumption fixes the authority boundary for the adversarial analysis: an attacker may exhaust enrichment capacity through the observed system trajectory, but STAK-PSAL itself does not become an actuator. A runtime-modifiable reduction operator introduces an attack surface absent in immutable certified pipelines; three risk classes follow from the enrichment structure (Section 2). *Spoofing* of the warning condition: an adversary perturbing the input near the safety boundary may force the monitor into META_REVIEW and consume its enrichment budget, raising $\mathcal{F}_n$ without any physical degradation, consistent with the adversarial elasticity inversion identified in Section 5. *Adversarial activation* of the enrichment pathway: a compromised channel may inject distortions into row 0 of W_t that survive the immediate admissibility check but corrupt downstream behaviour. *Destabilisation by repeated probing*: a sequence of bounded perturbations near the boundary can deplete the enrichment budget, driving the system toward the R2 regime even when each individual perturbation is benign. The attack surface is therefore not the physical process alone, but the finite enrichment capacity of the observational boundary itself. Horizon Debt sharpens this attack surface: an adversary need not force immediate rupture, but only consume enough repair horizon that authenticated enrichment mechanisms age out before they can arrive.

Adversarial exhaustion is more dangerous than direct drive to rupture: repeated bounded perturbations force META_REVIEW entry, consume enrichment capacity, and raise $\mathcal{F}_n$ while the physical process remains in bounds, producing adversarial elasticity inversion (Definition 7). Negative elasticity signals intrusion only under the compound condition of bounded $|\Delta P_t|$ and repeated META_REVIEW entry; 20% of R3 runs exhibit $\min E_r < 0$ in isolation (Table 5).

Mitigation is partial and deployment-dependent. Enrichment triggers should authenticate warning provenance; mutable pathways should be restricted to a rate-limited API surface; immutable variants preserve the monitoring contract without exposing the weight matrix. The security claim is deliberately bounded: STAK-PSAL does not identify an attacker from negative elasticity alone. It identifies a structural symptom that should not occur when the physical degradation proxy remains bounded and the enrichment pathway is being repeatedly exercised. A full adversarial analysis is a defined open boundary of the present work.

10 Related Work

Runtime verification and drift detection. Classical RV monitors traces against temporal specifications [1,2]; STAK-PSAL instead monitors the reduction operator M_t to anticipate violations. Unlike conventional drift monitors evaluating current observation space, STAK-PSAL evaluates the cumulative structural cost to maintain admissibility. Predictive monitoring [5,4] is the closest analogue, instantiated here for admissibility degradation via $\mathcal{F}_n$.

Monitorability and formal methods. The monitorability question [15,16] is here specialised to admissibility degradation, answering affirmatively for continuous trajectories. While static interpretation [12] and CEGAR [13] establish properties pre-deployment, STAK-PSAL monitors whether the runtime reduction preserves the requisite distinctions.

Cyber-physical systems and PHM. Runtime assurance for CPS [19,25] is extended to the observation layer. Prognostics and Health Management (PHM) tracks latent fatigue accumulation [26,24]; $\mathcal{F}_n$ specialises this to observational admissibility.

11 Applicability, Limitations, and Domain Boundaries

Model scope and statistical claims. The current prototype uses a linear-nonlinear reduction with two critical vectors and scalar degradation; multiple concurrent critical pairs (x_i, y_i) are the immediate scaling challenge, requiring fatigue aggregation or per-pair tracking. Results are point estimates on a fixed model with calibrated thresholds and require confidence quantification. The CRITICAL thresholds were set on the 13 deterministic runs of Section 7.2; the stochastic and benchmark sweeps are out-of-calibration evaluations, not cross-validation. The zero-false-negative claim is model-specific. Rapid degradation is no longer merely a reduced-lead-time limitation. It is represented by Horizon Debt: a state in which admissibility may still hold instantaneously, but timely admissibility has failed because the remaining rupture horizon is no greater than the monitor's lag. If no certified enrichment mechanism remains viable, the monitor cannot certify enrichment-mediated recovery. This does not prove physical inevitability of rupture; it proves only that recovery is no longer certified under the current mechanism catalogue and gain contracts.

Proof and runtime gap. The Coq theorems use Definition 2 (exact equality); the OCaml monitor uses Definition 3. The threshold-crossing bridge is machine-checked in `formal/threshold_bridge.v` in its existential form: under geometric contraction, crossing from any ε to any smaller ε' is guaranteed in bounded steps. Closing the remaining gap formally requires restating the structural invariants (T1)–(T5) over Definition 3; the explicit logarithmic ceiling bound of Lemma 1 remains a quantitative refinement of the mechanised existential.

Domain boundary: continuous fatigue mathematics and discrete generative systems. STAK-PSAL rests on assumptions fitting continuous degradation: multiplicative weight decay, a smooth separation margin, an additive enrichment response, and a continuous fatigue functional. These hold for cyber-physical processes; transfer to discrete generative systems (LLM agents and similar) is not clean: token-level policies do not exhibit multiplicative weight decay, the embedding analogue of separation margin is not continuous along a trajectory, and the operational meaning of enrichment is undefined. A possible analogue of enrichment would be adaptive resampling, tool-mediated verification, temperature reduction, or retrieval expansion after a warning signal; but none of these is a direct counterpart of restoring a physical separation margin. Applicability to discrete generative systems is therefore not an extension claimed here. It is a separate research problem: first define a monitorable discrete margin, then define what counts as finite recovery elasticity for that margin.

STAK-PSAL does not prove inevitability of rupture. The monitor predicts bounded rupture horizon under continued degradation dynamics within the evaluated model family. Sufficient external enrichment, operator intervention, or structural modification may still restore admissibility before collapse.

Conclusion

A runtime system may remain operationally closed while already possessing a bounded rupture horizon. The machine-checked semantic hierarchy (`formal/metastable_recovery.v`) formalises this as the admissible visibility failure theorem: operational closure does not entail structural safety, and metastable closure admits constructive rupture witnesses bounded within a declared horizon. The machine-checked existential bridge (`formal/threshold_bridge.v`) connects the strict-equality formal layer to the ε-admissible operational layer, guaranteeing threshold crossing under geometric contraction; the explicit logarithmic ceiling bound of Lemma 1 is the quantitative refinement isolating the precise horizon. STAK-PSAL operationalises these results: $\mathcal{F}_n$ tracks cumulative structural degradation invisible to instantaneous monitors, with CRITICAL signals providing mean lead time 11.6 to 13.7 steps and zero false negatives in the 2,159 evaluated ruptured runs; and the three-witness certificate emits a bounded rupture witness on confirmed collapse. The Metastable Recovery Principle identifies the risk static and output-level monitors cannot detect: $\mathcal{F}_n$ is 13.8–14.9× higher in metastably closed runs. Runtime assurance is therefore path-dependent rather than purely state-dependent: admissibility may hold at time t while the history required to preserve it still carries a bounded rupture horizon. Safety-relevant distinguishability must therefore be monitored as a runtime quantity. STAK-PSAL therefore separates three conditions that ordinary runtime status collapses: present admissibility, timely admissibility, and durable recovery. Structural fatigue records the historical cost of remaining admissible; Horizon Debt records the moment at which admissibility remains extensionally visible but operationally too late.

References

1. M. Leucker and C. Schallhart. A brief account of runtime verification. *Journal of Logic and Algebraic Programming*, 78(5):293–303, 2009.
2. K. Havelund and D. Peled. Runtime verification: From propositional to first-order temporal logic. In *International Conference on Runtime Verification*, pages 90–112. Springer, 2018.
3. S. A. Seshia, D. Sadigh, and A. D. Dragan. Toward verified artificial intelligence. *Communications of the ACM*, 61(7):46–55, 2018.
4. F. Chen and G. Roşu. Parametric trace slicing and monitoring. In *TACAS*, pages 246–261. Springer, 2009.
5. M. Zhao and G. Rozier. Predictive runtime monitoring for linear stochastic systems. In *International Conference on Runtime Verification*, pages 382–400. Springer, 2021.
6. E. Kim, M. Guo, and B. Fidan. Online fault prognosis for cyber-physical systems. *IEEE Transactions on Automation Science and Engineering*, 2022.

7. C. E. Tuncali, T. P. Pavlic, and G. Fainekos. Simulation-based adversarial test generation for autonomous vehicles. In *IEEE Intelligent Vehicles Symposium*, 2018.
8. D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané. Concrete problems in AI safety. *arXiv:1606.06565*, 2016.
9. M. Scheffer *et al.* Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009.
10. C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4(1):1–23, 1973.
11. S. L. Pimm. The complexity and stability of ecosystems. *Nature*, 307(5949):321–326, 1984.
12. P. Cousot and R. Cousot. Abstract interpretation: A unified lattice model. In *POPL*, pages 238–252, 1977.
13. E. Clarke *et al.* Counterexample-guided abstraction refinement. In *CAV*, pages 154–169. Springer, 2000.
14. A. Pnueli. The temporal logic of programs. In *FOCS*, pages 46–57. IEEE, 1977.
15. A. Francalanza, J. A. Pérez, and C. Sánchez. Runtime verification for decentralised and distributed systems. In *Lectures on Runtime Verification*, pages 176–210. Springer, 2018.
16. A. Bauer, M. Leucker, and C. Schallhart. The good, the bad, and the ugly, but how ugly is ugly? In *International Conference on Runtime Verification*, pages 126–138. Springer, 2011.
17. Y. Falcone, J. J. Hadjadj-Aoul, and T. Jéron. More lessons on test generation for temporal properties. In *International Workshop on Formal Methods for Testing*, pages 126–145. Springer, 2010.
18. E. A. Lee and S. A. Seshia. *Introduction to Embedded Systems: A Cyber-Physical Systems Approach*. MIT Press, second edition, 2017.
19. S. A. Seshia and S. Jha. Adaptive runtime monitoring for autonomous cyber-physical systems. In *17th International Conference on Hybrid Systems: Computation and Control*, pages 173–182. ACM, 2014.
20. M. Viswanathan and R. Vardi. Revisiting the hierarchy of until and since. In *Logic in Computer Science (LICS)*, pages 70–79. IEEE, 1994.
21. A. M. Saxe, Y. Barda, and T. B. Bright. Mitigating adversarial effects through randomization. In *International Conference on Learning Representations*, 2019.
22. H. T. Pham, A. Khatkhate, Y. Zhang, and Z. Gao. Probabilistic approach to remaining useful life estimation. *Annual Reviews in Control*, 39:133–142, 2015.
23. Y. Li, N. Chen, J. Xu, and X. Sheng. Degradation data analysis based on a wiener process with multiple change-points and measurement error. *European Journal of Operational Research*, 271(2):641–652, 2018.
24. L. H. Chiang, E. L. Russell, and R. D. Braatz. *Fault Detection and Diagnosis in Industrial Systems*. Springer Science+Business Media, 2001.
25. J. D. Schierman, D. G. Ward, N. D. Richards, N. Gandhi, J. F. Hammer, and D. B. Murray. Runtime assurance for advanced flight-critical systems. *IEEE Transactions on Aerospace and Electronic Systems*, 51(4):2931–2945, 2015.
26. A. K. S. Jardine, D. Lin, and D. Banjevic. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7):1483–1510, 2006.